



MesoProbe

THE NEXT GENERATION OF
**MESO-SCALE
MECHANICAL TESTING**



NANOMETER
PRECISION



INTEGRATED
DIC

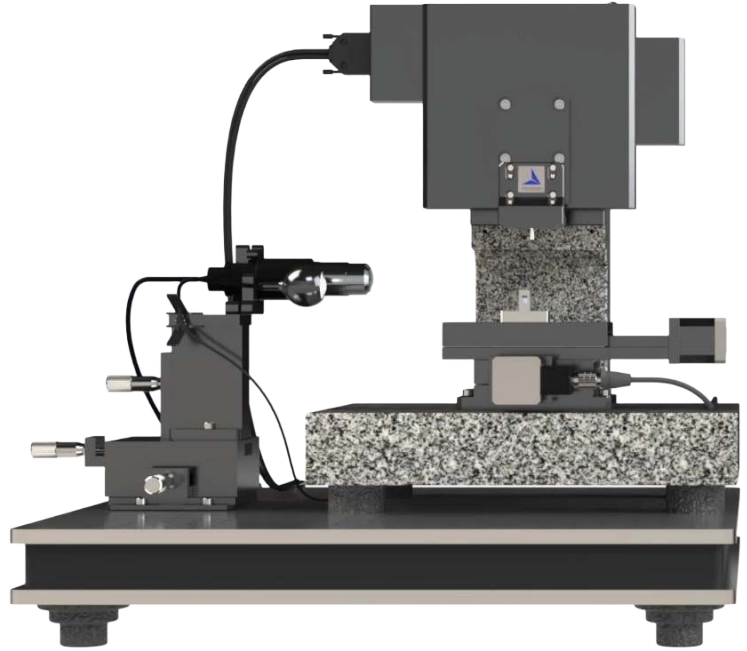


HIGH
THROUGHPUT

WHY CHOOSE

MesoProbe

MesoProbe is designed to bridge the gap between nano and macro testing—enabling accurate mechanical characterization on small samples with bulk-relevant insights.



THE GAP IN MECHANICAL TESTING



NANO SCALE

High resolution
Very small test volumes
Limited representation of
bulk behavior

1 nm – 10 μ m



MESO SCALE

The Sweet Spot

Bridges micro and macro
Small sample volume
High-throughput data
Integrated DIC
High-temperature testing
Multiple test modes

10 μ m – 5 mm



MACRO SCALE

Bulk properties
Large sample volume
Low spatial resolution

> 5 mm



HOW IT WORKS



01

Small Sample
Minimal material
requirement



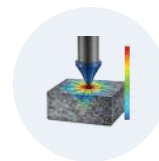
02

Mechanical Loading
Indentation, Compression,
Tensile, or Bending



03

High Resolution Imaging
Real-time in-situ optical
observation



04

Digital Image Correlation
Full-field strain mapping
with high accuracy



05

Mechanical Properties
Stress, Strain, Modulus,
Creep, Fatigue & more

ONE PLATFORM. MULTIPLE EXPERIMENTS.

MesoProbe supports a wide range of mechanical tests with integrated Digital Image Correlation for full-field strain analysis and high-throughput data generation.

EXPERIMENTS

1. INDENTATION



Indentation Test using a Spherical Tip

- Hardness
- Elastic Modulus
- Load-Displacement Analysis
- Depth Sensing Indentation

2. COMPRESSION



Compression Test on a Soft Material Matrix

- Polymers, Hydrogels, Rubber, Foams
- Soft Materials Testing
- Low Force Sensitivity (μN Level)

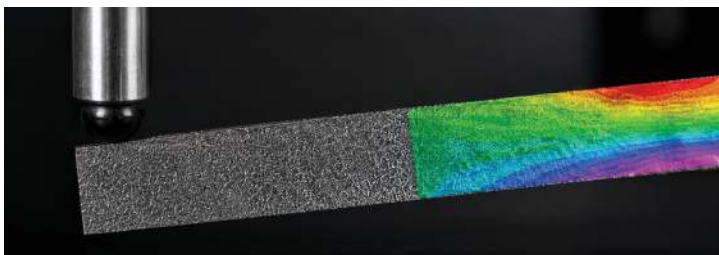
3. THREE-POINT BENDING



Three Point Bending Test on Soft Material Matrix with Strain Data from DIC Analysis data Overlayed

- Young's Modulus
- Stress & Strain Mapping
- Full-field Strain from DIC
- Fracture & Failure Analysis

4. CANTILEVER BENDING



Cantilever Bending Test on Aluminium Sample with Strain Data from DIC Analysis data Overlayed

- Creep & Fatigue Analysis
- Strain Evolution
- High Throughput Creep Testing
- Ideal for Limited Sample Material

INDUSTRIES & RESEARCH AREAS



AUTOMOTIVE



AEROSPACE



BATTERY MATERIALS



THIN FILMS & COATINGS



SEMICONDUCTORS & MEMS



BIOMATERIALS & MEDICAL DEVICES



EDUCATION & RESEARCH LABS

TECHNICAL SPECIFICATIONS

SPECIFICATION	DETAILS
 Maximum Actuation Load	20 N
 Maximum Displacement	60 mm
 Displacement Resolution	1 nm
 Camera Resolution	4024 px × 3036 px
 Temperature Capability	Up to 600°C
 Motorized Translation Stages	X-axis: 150 mm Y-axis: 50 mm
 Optics	0.2X (1X / 5X / 10X Optional)



SMALL SAMPLES.
BIG INSIGHTS.

TEST MODES



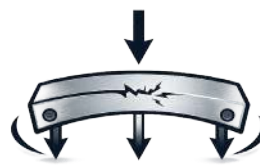
FRACTURE



FATIGUE



CREEP



BENDING



TENSILE

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Scan to Explore
MesoProbe
 in Action